Module: CSCI369 Ethical Hacking
Assignment: Assignment Q3
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Concepts:

Question 3 uses a client-server architecture over TCP connection. This implementation is called the **reverse shell**. The client initiates and connects to the server instead of the server connecting to the client.

How does it works?
Server (server.py): This is the operator side that listens on port 4444 and accepts an incoming connection from that port.
Client (revshell.py): This is the file that was supposedly run on the victim's machine by the victim. When executed, it initiates a connection back to the server.

"cd" command handling:

- With the help os.chdir(), I was able to make the program maintain a persistent "current directory" by changing the Python process directory.
- subprocess.Popen() was used instead of subprocess.ran(). subprocess.Popen() supports the use of cwd=... to set the working directory of the spawned process.

In this assignment, both server.py and revshell.py will be running on the same virtual machine as per specified in the assignment brief.

Prerequisites:

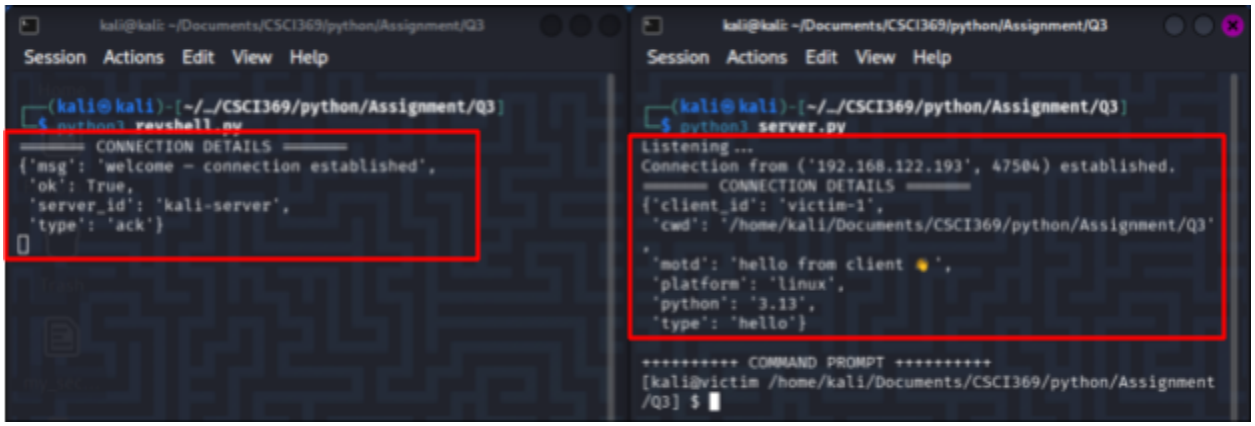
- Navigate into the directory with both your server.py and revshell.py are located.

Execution:

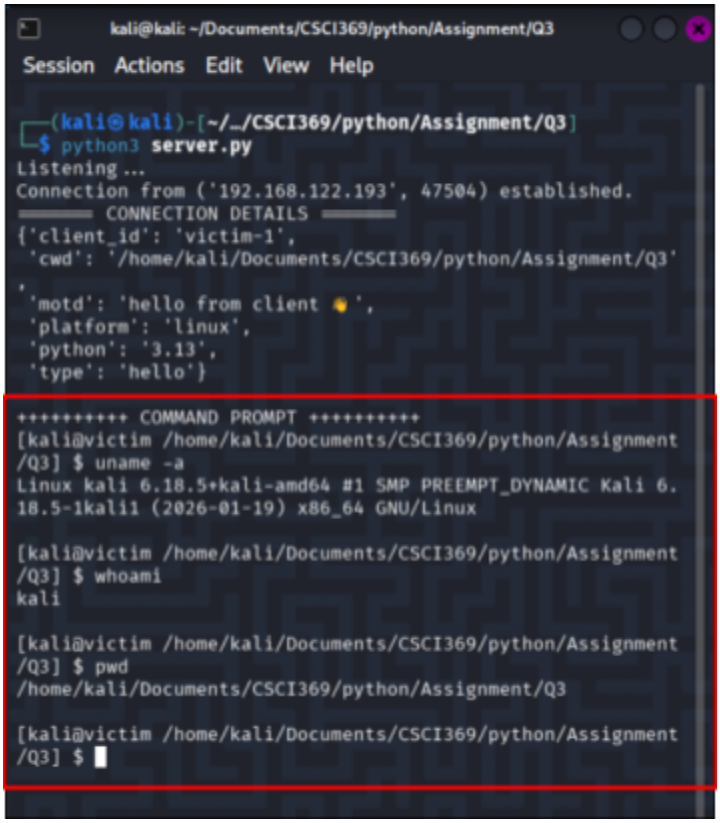
Command to run code:

python3 revshell.py
python3 server.py

[Kali] Run each file in a separate terminal. The server and client will do a simple handshake to confirm their connection.



[Kali] Ran commands: uname -a, whoami, pwd to show that we are connected to the client. Look at the output returned.



[Kali] Program performing cd commands.

- (Purple) Change in working directory path
- (Red) After changing directory
- Directory path is persistent

```
++++++++++ COMMAND PROMPT ++++++++++
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ uname -a
Linux kali 6.18.5+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.
18.5-1kali1 (2026-01-19) x86_64 GNU/Linux

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ whoami
kali

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ pwd
/home/kali/Documents/CSCI369/python/Assignment/Q3

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ cd ..

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
] $ ls
Q1
Q2
Q3
Q4
Q5

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
] $ pwd
/home/kali/Documents/CSCI369/python/Assignment

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
] $
```

[Kali] Program performing cd commands.

- (Red) Change in working directory path
- (Turquoise) Change in folder contents
- Directory path is persistent

```
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ pwd
/home/kali/Documents/CSCI369/python/Assignment/Q3
```

```
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ ls
revshell.py
server.py
```

```
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ cd /home/kali/Documents/CSCI369/txtFiles
```

```
[kali@victim /home/kali/Documents/CSCI369/txtFiles] $ ls
cipher.txt
decdata.txt
mypassword.txt
openssl_plain.txt
rev_shell.txt
```

```
[kali@victim /home/kali/Documents/CSCI369/txtFiles] $ pwd
/home/kali/Documents/CSCI369/txtFiles
```

```
[kali@victim /home/kali/Documents/CSCI369/txtFiles] $
```

[Kali] Program performing cd commands.

- (Red) Change in working directory path
- (Turquoise) Change in folder contents
- Directory path is persistent

```
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ cd /home/kali/Documents/CSCI369/txtFiles

[kali@victim /home/kali/Documents/CSCI369/txtFiles] $ ls
cipher.txt
decdata.txt
mypassword.txt
openssl_plain.txt
rev_shell.txt

[kali@victim /home/kali/Documents/CSCI369/txtFiles] $ pwd
/home/kali/Documents/CSCI369/txtFiles

[kali@victim /home/kali/Documents/CSCI369/txtFiles] $ cd /h
ome/kali/Documents/CSCI369/python/Assignment/Q2

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q2] $ ls
myvenv
paramiko_SSH.py
passwordFile.txt
pwFile.txt

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q2] $ pwd
/home/kali/Documents/CSCI369/python/Assignment/Q2

[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q2] $
```

Execution:

To end connection, use either of the following options:

- Ctrl + c
- exit
- quit

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q3]
└─$ python3 revshell.py
====== CONNECTION DETAILS ======
{'msg': 'welcome — connection established',
 'ok': True,
 'server_id': 'kali-server',
 'type': 'ack'}
^C
[*] Client exiting (Ctrl+C)...
```

```
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q2] $ ^C
[*] Exiting (Ctrl+C)...
```

[Kali] Typing ‘exit’ / ‘quit’ in server.py, exits both server.py & revshell.py

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q3]
└─$ python3 server.py
Listening...
Connection from ('192.168.122.193', 38898) established.
====== CONNECTION DETAILS ======
{'client_id': 'victim-1',
 'cwd': '/home/kali/Documents/CSCI369/python/Assignment/Q3'
,
 'motd': 'hello from client 👋',
 'platform': 'linux',
 'python': '3.13',
 'type': 'hello'}

++++++++++ COMMAND PROMPT ++++++++++
[kali@victim /home/kali/Documents/CSCI369/python/Assignment
/Q3] $ exit
[*] Exiting (operator requested)

┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q3]
└─$
```